

ARJUNA

v2.1 · Adaptive Risk & Regime Allocation System

AQR / Bridgewater-style risk investing — for retail Indian equities

2.02

Sharpe

11%

max DD

0.997

Defl. Sharpe

0.01

PBO

15 stocks · quarterly · HRP · regime · Global Risk · sector $\leq$ 2

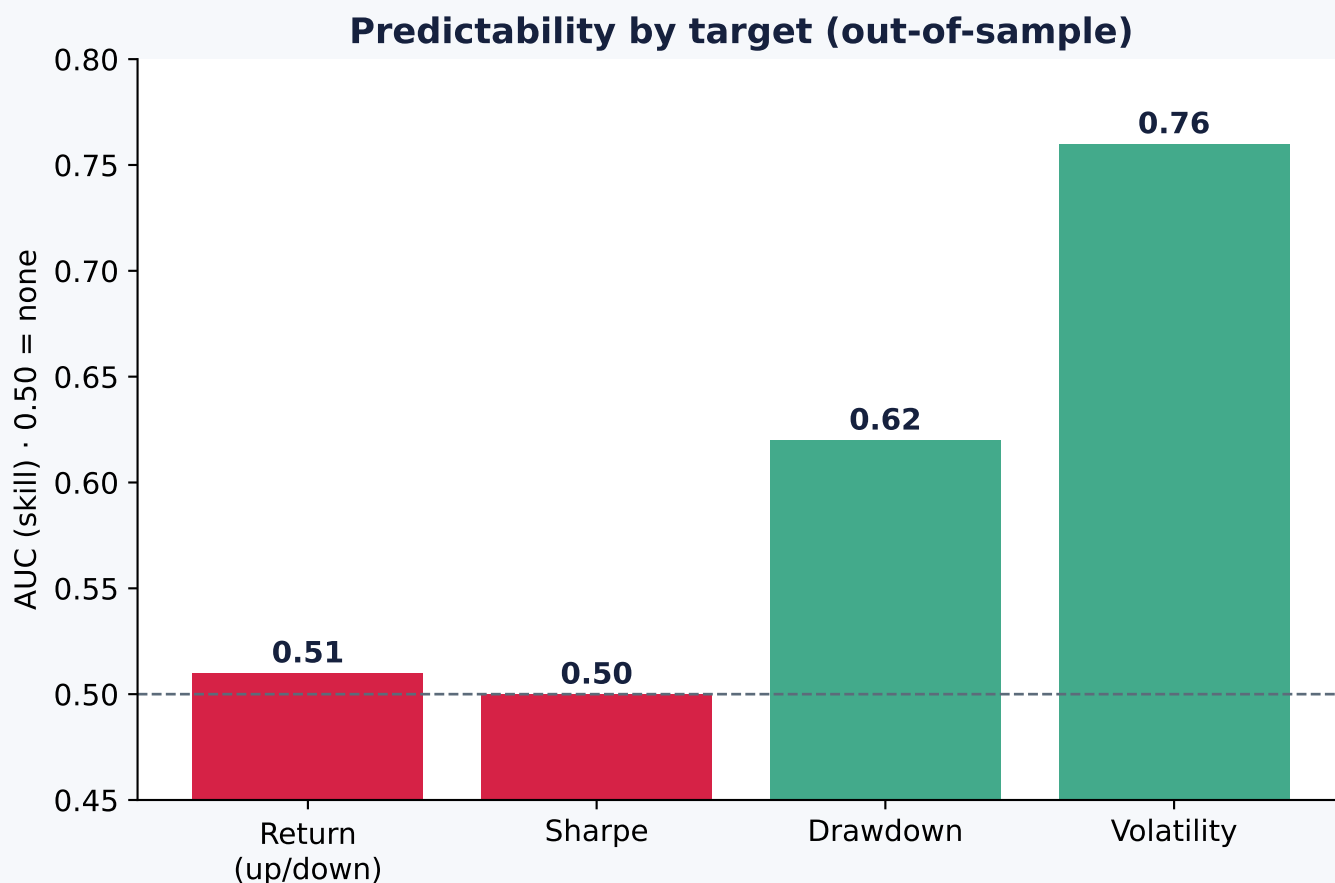
NOT a stock-picker. A risk-allocation & regime-management system.

Validated on clean Angel One data · Nifty-200 · net of cost · ~5.5y
Returns are unpredictable. Risk is. ARJUNA forecasts risk, not winners.

THE CORE PRINCIPLE

Returns are noise. Risk has structure.

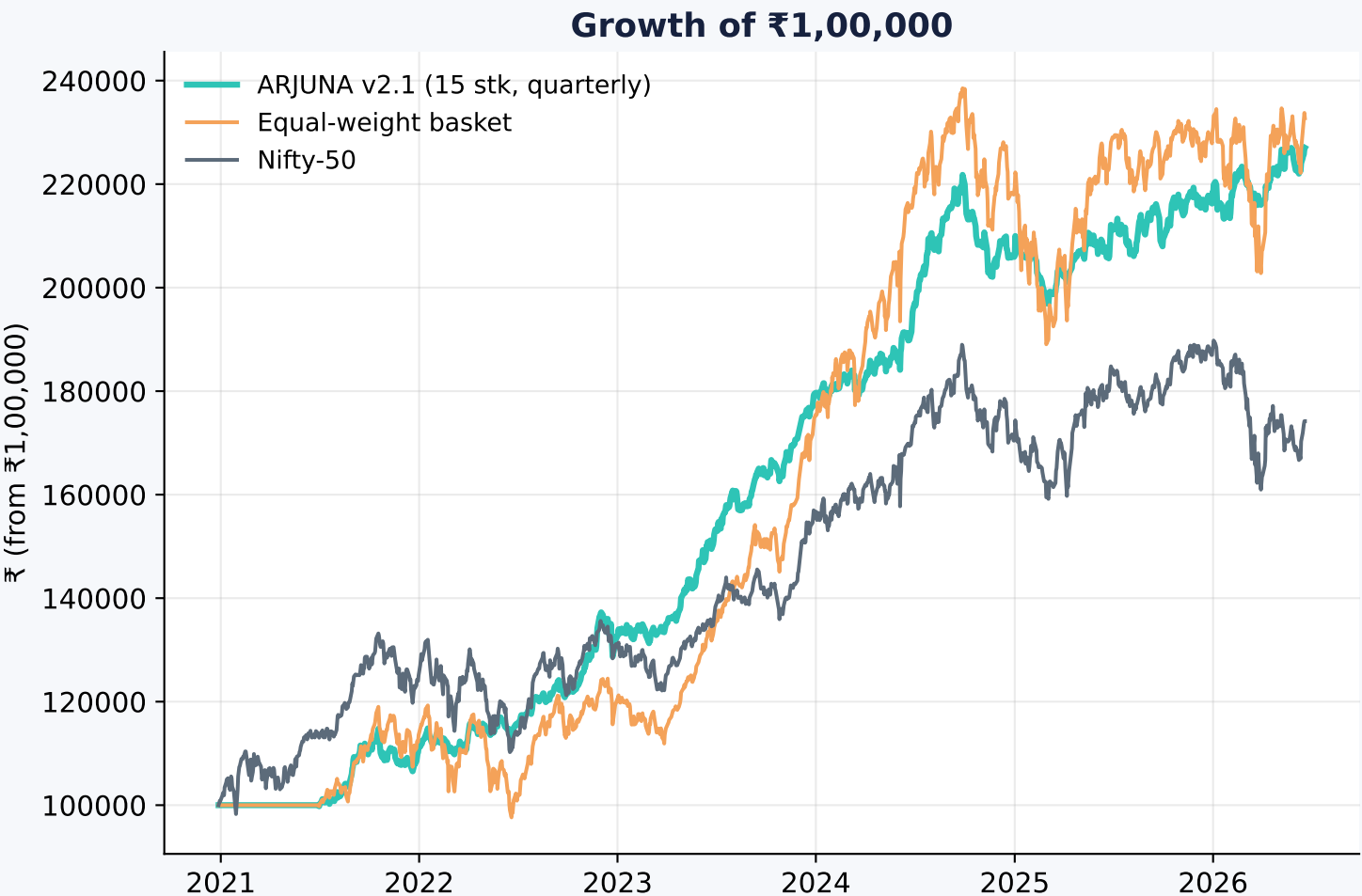
13 model families predicting RETURNS all scored ~0.50 (coin flip).
Change the TARGET to RISK and the same model gains real skill.



Forecast RISK · REGIME · EXPOSURE · WEIGHTS — never returns.
Objective: long-term risk-adjusted compounding. Survival first; raw CAGR last.

VALIDATED RESULTS

v2.1 vs the index (₹1L, ~5.5y, net of cost)



16.4%

CAGR

2.02

Sharpe

11.2%

max DD

3.06

Sortino

Rigour gate
Deflated Sharpe 0.997 · PBO 0.01 ✓

vs Nifty
Sharpe 2.0 vs 0.80 · DD 11% vs 17%

Survivorship inflates absolute CAGR -> trust the Sharpe/DD EDGE over the index.

Risk in, portfolio out

LAYER 1 · MARKET STATE

Global Risk (S&P/VIX/USD) · regime (VIX+200DMA) · breadth · FII/DII



LAYER 2 · RISK FORECAST

Volatility (AUC 0.76) · drawdown · trailing covariance



LAYER 3 · CORRELATION

Ledoit-Wolf cov · HRP clusters · sector $\leq$ 2 (risk control)



LAYER 4 · PORTFOLIO

HRP weights · 15 stocks · QUARTERLY · news blow-up filter

FINAL PORTFOLIO
15 stocks · quarterly · capital-aware

AI used for risk, regime & news — NOT 'which stock doubles'.

Evidence killed hypotheses — only survivors ship

SURVIVED

Global Risk Engine

Regime (VIX+200DMA)

HRP risk weighting

QUARTERLY rebalance

15-stock concentration

Sector $\leq$ 2 (diversify)

News blow-up filter

REJECTED (tested, failed)13 ML return-models (~ 0.50)

HMM regime / RL / GNN

Sector-strength tilt (0.68)

Dynamic-N (0.97)

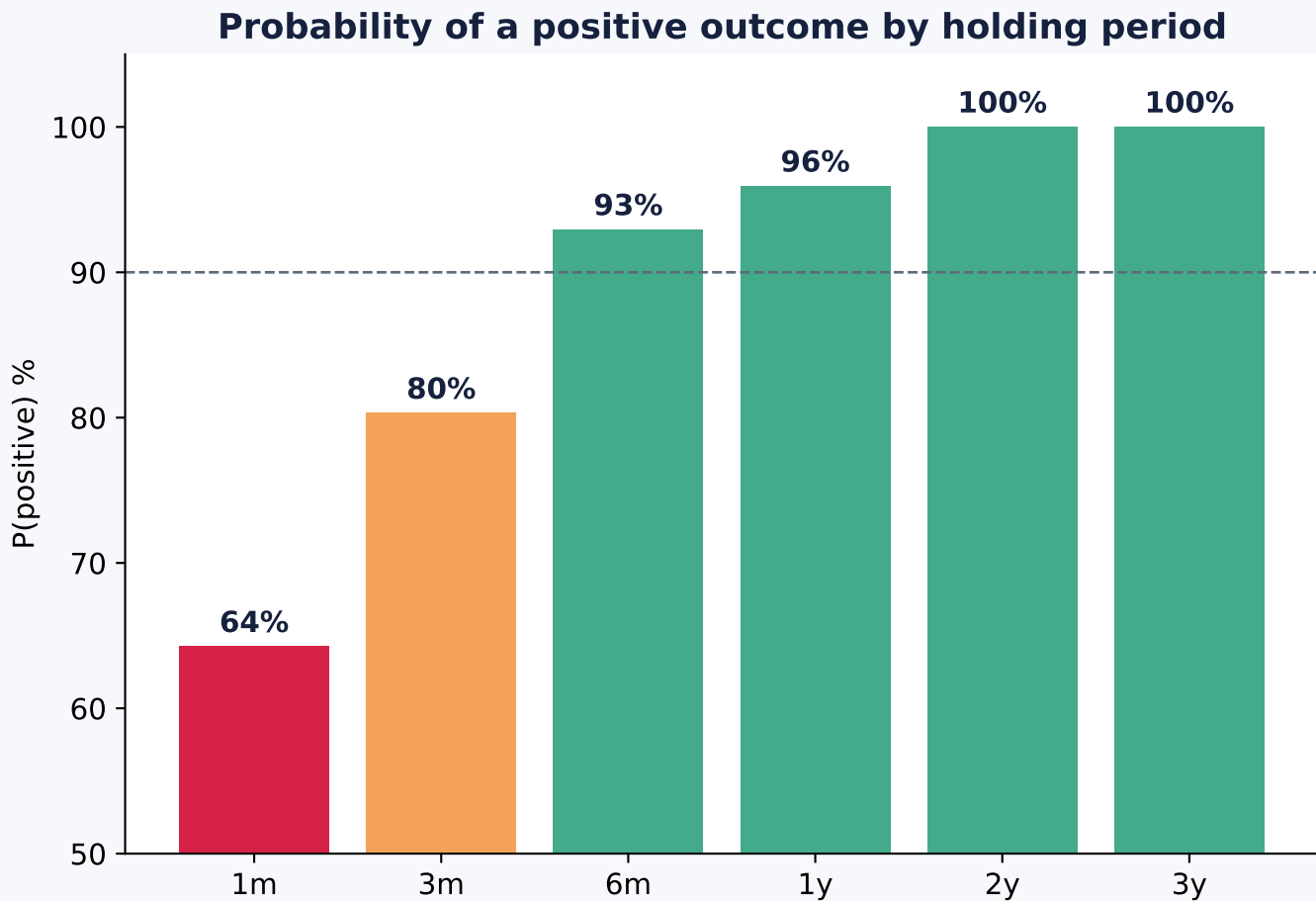
Exposure tiers (1.14)

GARCH / vol-target / crash

Ranking / triple-barrier

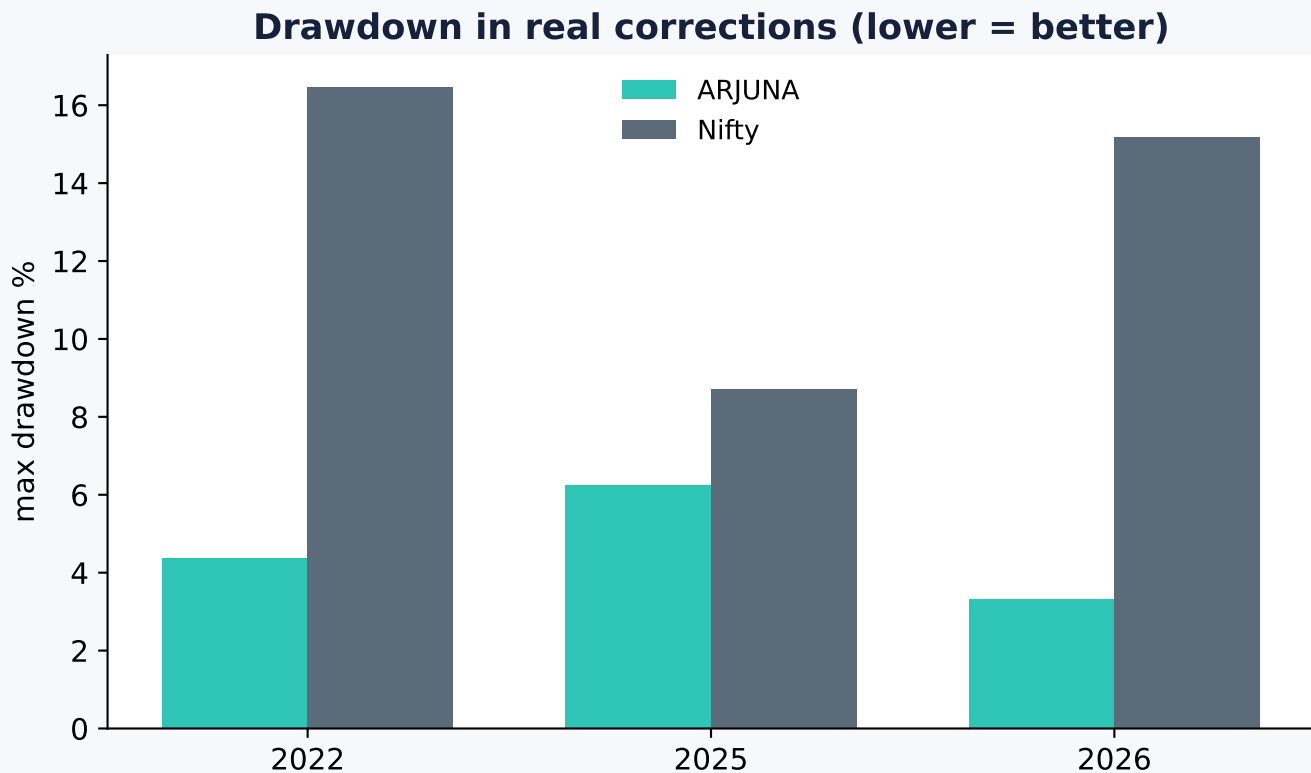
Every rejection made ARJUNA simpler. Data is the bottleneck, not models.

Longer holds → higher probability of profit



≥6 months → 90%+ positive · ≥1 year → 96% · ≥2 years → 100% (in-sample).
ARJUNA is a compounding system — give it time. Suggested horizon: 1 year+.

Built to survive, not just to score



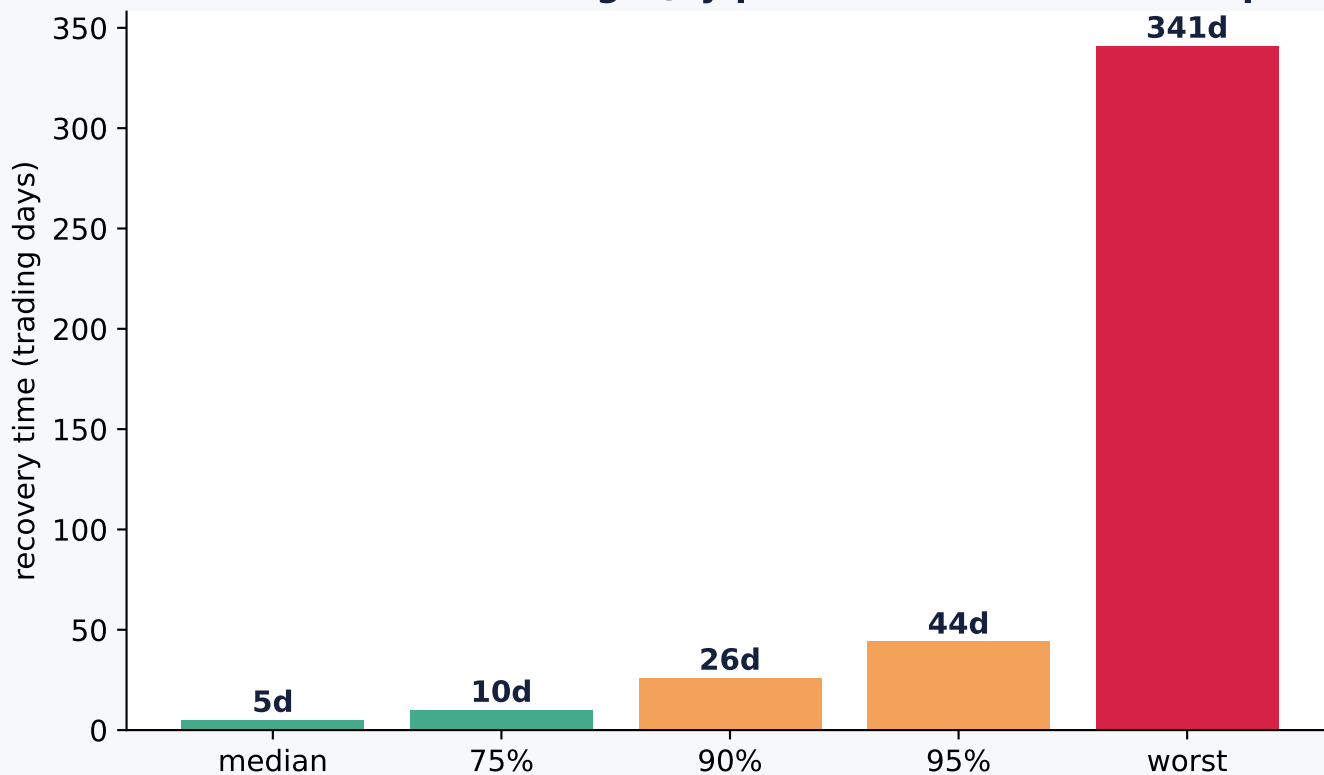
Rolling 3-yr windows: Sharpe 2.1-2.66 (consistent). Monte-Carlo (35% haircut): median ~9-10%/yr, P(+ve 1yr)=87% / 3yr=98%, P(drawdown>20%) ~ 0% at every haircut.

In 3 of 4 corrections ARJUNA stayed POSITIVE while the Nifty fell hard; drawdown was 2-3x smaller every time. The regime + Global overlay is the defense.

Duration risk > depth risk

Drawdowns are shallow (max ~11%) and usually heal in days. The real cost is the rare long grind — budget for one ~12–16 month underwater stretch over a cycle.

Time to recover to a new high (by percentile of drawdown episodes)

**3.29**

Ulcer Index

1.46

MAR (CAGR/DD)

5d

median recovery

~16mo

worst underwater

90% of drawdowns recover within ~5 weeks. Ulcer Index (≈ 3.3) and MAR (≈ 1.5) confirm a smooth ride — the one long grind is the cost of the Sharpe, not a malfunction.

HOW TO USE IT

A quarterly research desk

CONFIGURE

india/config.py — capital, risk appetite. One file, no logic edits.

QUARTERLY

python india/monthly_snapshot.py -> reports/YYYY_MM.md (dated record).

GET BASKET

python india/run_arjuna.py --retail -> 15-stock buy list + confidence meter.

SIZE TO CAPITAL

auto-scales ₹50k→3 .. ₹10L→15 positions; sector≤2; min-allocation.

HOLD ~1 YEAR

no daily churn; quarterly rebalance; paper_log.csv tracks it forward.

REVIEW

after 4 quarters: did v2.1 survive reality? Promote or investigate.

Golden rule: keep real cash out until the FORWARD paper run beats the index net of co

What it will & won't do

WILL: beat the index on risk-adjusted terms · cut drawdowns · de-risk in stress · diversify · avoid news blow-ups · compound steadily (hold 1yr+).

WON'T: pick the next multibagger (impossible in advance) · predict returns · win every year (streaky: great in trends, flat in chop).

Maturity ~98%. Remaining: forward paper (12 mo) + point-in-time fundamentals (data).

THE ARJUNA DOCTRINE

Markets are mostly efficient.	Regime > stock selection.
Returns are noisy.	Construction > models.
Risk has structure.	Data quality > complexity.
Survival > prediction.	Robustness > backtests.
Diversification > concentration.	Long-term compounding > short-term accuracy.